Energy Equations for Part 1

Kinetic Energy (K)

For a point mass m at the end of a rod of length ℓ rotating with angular velocity $\dot{\theta}$: - Velocity magnitude: $v = \ell\dot{\theta}$ - Kinetic Energy: $K = \frac{1}{2}mv^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$

Potential Energy (P)

The total potential energy is the sum of gravitational potential energy and the energy stored in the nonlinear spring.

Gravitational Potential Energy (V_g)

Relative to the joint (origin): - $y = \ell\sin(\theta)$ - $V_g = mgy = mg\ell\sin(\theta)$

Nonlinear Spring Potential Energy (V_{spring})

As given in the problem: - $V_{spring} = \frac{1}{2}k_1\theta^2 + \frac{1}{4}k_2\theta^4$

Total Potential Energy

- $P = V_g + V_{spring} = mg\ell\sin(\theta) + \frac{1}{2}k_1\theta^2 + \frac{1}{4}k_2\theta^4$

Lagrangian (L)

- $L = K - P$
- $L = \frac{1}{2}m\ell^2\dot{\theta}^2 - \left(mg\ell\sin(\theta) + \frac{1}{2}k_1\theta^2 + \frac{1}{4}k_2\theta^4\right)$